INCH-POUND

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SUPERSEDING

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DETAIL SPECIFICATION

SIGNAL KIT, PERSONNEL, DISTRESS, A/P25S-5B



Comments, suggestions, or questions on this document should be addressed to: AFLCMC/EBH Munitions Sustainment Division, 6033 Elm Lane, Bldg. 1247, Hill AFB UT 84056-5819 or email to: AFLCMC.EBH.MunitionsWorkflow@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE.

1.1 Scope. This specification covers the performance requirements for one (1) type of airborne personnel distress signal kit designated A/P25S-5B and consisting of a hand-fired launcher and a bandoleer assembly containing seven (7) red signals.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections three (3) and four (4) of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet the specified requirements documents cited in sections three (3) and four (4) of this specification, whether or not they are listed.

2.2 Government Documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation of contract. (see 6.2)

MIL-STD-1916	DoD Preferred Methods for Acceptance of Product
MIL-STD-331	Department of Defense Test Method Standard

(Copies of these documents are available online at https://assist.dla.mil.)

2.3 Non-Government publications. N/A

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

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3.3 Workmanship. The signal kit, including all parts and equipment, shall be constructed in accordance with commonly accepted industrial workmanship standards. All components of a signal kit shall be functional without sharp edges, burrs, or protrusions.

3.4 Materials. All materials shall be chosen or suitably treated to resist corrosion due to electrolytic decomposition, fungus, salt spray, and any other atmospheric condition that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals (ODC) shall be avoided, whenever feasible.

3.5 Performance.

3.5.1 Function. The signal kit when operated at ambient conditions, shall function as follows:

a. A launcher shall require the use of only one (1) hand to fire a signal.

b. The fit of the signal-gripping device of the launcher with a signal shall preclude the signal from falling out of the launcher when the loaded launcher is aimed vertically at the ground.

c. No more than two (2) attempts shall be permitted to initiate the signal, except when the signal has been subjected to low temperature (see 3.8.1.1 and 3.8.3.3) in which case no more than four (4) attempts shall be permitted.

d. The signal shall not be held captive in the launcher while the rocket motor burns.

e. The signal shall not explode or deflagrate, or the components separate or come apart, within 15-feet of the launcher.

f. Venting of the rocket motor resulting in snuffing out of the propellant system after initial ignition shall not occur.

g. The ignition of the candle shall not occur below 400-feet above ground level when fired from a vertically aligned launcher.

h. The minimum burn time and apogee of each display (when fired from a vertically aligned launcher with a wind speed not greater than 10-miles per hour) shall be as specified in Table I. The candle shall burn through apogee and burn out prior to hitting the ground. Non-ignition of the display shall be considered as zero burn time.

i. The apogee of the display shall not deviate more than 30° (degrees) from vertical when fired from a vertically aligned launcher with a wind speed not greater than 10-miles per hour.

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3.5.2 Burn time, candlepower, and color.

3.5.2.1 Burn time. The candle of the signal shall produce a display that meets the following requirements:

Min Individual Burn Time (sec)	Min. Average Burn Time (sec)
8.0	9.0

3.5.2.2 Candlepower. The candle of the signal shall produce a red display that has a minimum luminous intensity of 26,000 lumen seconds and a minimum average candlepower value of 2,500 lumens for the first seven (7) seconds.

3.5.2.3 Color. The candle of the signal shall produce a red display that has a dominant wavelength of 620 ± 20 millimicrons. The purity of the color of the red display shall be a minimum of 76% (percent).

3.5.3 Safety. A signal with no visible sign of damage shall not malfunction in a hazardous manner when operated (see 6.5). A signal with a visible sign of damage shall be safe to handle for disposal.

3.5.4 Penetration. The signal, when fired, shall be able to completely penetrate moderately dense jungle or forest foliage and produce a display.

3.5.5 Jolt. The signal shall withstand at least 1,750 impacts in three (3) axially perpendicular orientations consisting of half sine wave pulses having 230g peak acceleration for 2.0 milliseconds and a pulse rate of 35 impacts per minute without burning, exploding, or becoming unsafe to handle. The signal shall not malfunction in a hazardous manner when fired or handled following these impacts (see 6.5).

3.5.6 Jumble. The signal shall withstand repeated random impacts when rotated in a closed box at a speed of 30 ±2 revolutions per minute (rpm) for a total of 3600 ±10 revolutions without burning, exploding, or becoming unsafe to handle. The signal shall not malfunction in a hazardous manner when fired or handled following these impacts (see 6.5).

3.5.7 Forty-foot drop. The signal shall withstand a 40-foot drop without burning, exploding, or becoming unsafe to handle. The signal shall not malfunction in a hazardous manner when fired or handled following this drop (see 6.5).

3.5.8 Altitude. The signal shall function as specified in 3.5.1 after exposure to atmospheric pressure at an altitude of 50,000 feet for five (5) minutes.

3.5.9 Water immersion. The signal shall function as specified in 3.5.1 after being immersed in water for three (3) hours.

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3.5.10 Firing pin force (launcher). The firing pin force of the launcher shall be sufficient so that the launcher will initiate the signal primer. The firing pin force shall be a minimum of six (6) pounds.

3.6 Item identification. Each signal, launcher, and bandoleer shall be marked for identification (see 6.2). Minimum marking requirements defined in the Technical Data Package (TDP).

3.7 Interchangeability. All parts having the same Part Number (P/N) within the same lot shall be functionally and dimensionally interchangeable.

3.8 Environmental requirements.

3.8.1 Temperature and humidity. The signal shall function as specified in 3.8.1.1 and 3.8.1.2, after being subjected to high and low temperature variations (-65° to +160°F) for a period of 14 days with at least 95% relative humidity.

3.8.1.1 Low temperature. The launcher shall fire and the signal shall function as specified in 3.5.1, except for the display burn time and apogee altitude as listed in TABLE I, after the signals and launcher have been conditioned to -65° ±5°F for five (5) hours and immediately fired.

3.8.1.2 High temperature. The launcher shall fire and the signals shall function as specified in 3.5.1, except for the display burn time and apogee altitude as listed in TABLE I, after the signals and launcher have been conditioned to 160° ±5°F for five (5) hours and immediately fired.

3.8.2 Salt spray (launcher). The launcher shall fire a minimum of seven (7) signals without requiring maintenance after being exposed to a five (5)% salt concentration of salt spray for a minimum of 50 hours. Wiping off visible salt residue with a cloth in the collet area is allowable.

3.8.3 Storage.

3.8.3.1 Extreme high temperature storage. The signal shall withstand storage at a temperature of 200° ±5°F for a 14-day period without burning, exploding, or becoming unsafe to handle. The signal shall not malfunction in a hazardous manner when function tested at ambient temperature (see 6.5) following this storage. Note: Not required to meet requirements of 3.5.1 following this test.

3.8.3.2 High temperature storage. The signal shall function as specified in 3.5.1, except for the display burn time and apogee altitude as listed in TABLE I, when it has been stored at 160° ±5°F for 28-days.

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3.8.3.3 Low temperature storage. The signal shall function as specified in 3.5.1, except for the display burn time and apogee altitude as listed in TABLE I, when it has been stored at -65° ±5°F for 28 days.

3.8.4 Vibration. The signal kit shall withstand a vibration of ten (10) to 60 Hz, with amplitude maintained at 0.10 ±0.01 in. double amplitude up to and including 20 Hz and at two (2) ±0.2g peak for the remaining frequency coverage. The signal kit shall then function as specified in 3.5.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on signal kits representative of the production units to be supplied by the vendor. The first article tests shall include the tests and quantities specified in TABLE II.

4.3 Conformance inspection. Conformance inspection shall include the visual examination in 4.6.1 and the sample tests in 4.3.1. Lot size shall be a maximum of 5,000 kits (35,000 signals). Samples required for test purposes may be in addition to the deliverable units.

4.3.1 Sample tests. The conformance sample plan shall include the examinations, tests and quantities specified in Table III.

4.4 Test conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method documents or applicable paragraphs in this specification. Suggestions for certain test procedures are provided in section 6.4.

4.5 Requirements cross-reference matrix. TABLE IV provides a cross-reference matrix of the section three (3) requirements tested or verified in the paragraphs below.

4.6 Tests.

4.6.1 Visual inspection. The signal kit shall be visually inspected to determine compliance with section three (3) requirements in particular, this shall include the requirements of 3.2, 3.3, 3.4, 3.5.3, and 3.7.

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4.6.2 Performance tests.

4.6.2.1 Function test. The function of the signal kit shall be determined. Only launchers and flares meeting the safety inspection criteria in 4.6.1 shall be used for this test. The launcher shall be held for firing with only one hand, signal end up in a true vertical position, and a minimum of 12-inches above the head, or the launcher shall be fixed for remote firing in a suitable test fixture.

a. After loading the signal into the launcher, the loaded launcher shall be aimed vertically at the ground to ensure that the signal does not fall out of the launcher.

b. A maximum of two (2) attempts to initiate the signal shall be made, except that a maximum of four (4) attempts shall be made with signals that have been subjected to low temperature (see 3.8.1.1 and 3.8.3.3).

c. When the signal is fired, it shall not be held captive in the launcher while the rocket motor burns.

d. When the signal is fired, it shall not explode or deflagrate, and components of the signal shall not separate or come apart within 15 feet of the launcher.

e. The rocket motor shall not vent such that the propellant system is snuffed out after initial ignition.

f. The height at which the candle ignites shall be in accordance with TABLE I.

g. The burn time of the display shall be in accordance with TABLE I.

h. The apogee altitude of the display shall be in accordance with TABLE I.

i. The angular deviation from vertical of the signal at apogee shall not exceed 30° (degrees) with wind conditions of 10-miles per hour or less.

j. The candle shall burn out prior to hitting the ground.

4.6.2.2 Burn time, candlepower, and color tests. Tests for burn time, candlepower, and color shall verify the performance of the candle only and can be accomplished by igniting the candle in a lab environment

4.6.2.2.1 Burn time test. The burn time of the candle shall be in accordance with 3.5.2.1.

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4.6.2.2.2 Candlepower test. The minimum luminous intensity and minimum average candlepower of the candle shall be 26,000 lumen seconds and 2,500 lumens, respectively (see 6.4.1).

4.6.2.2.3 Color test. The dominant wavelength and the purity of color of the candle shall be 620 ±20 millimicrons with a purity of 76% (see 6.4.1).

4.6.2.3 Penetration test. The penetration capability of the signal through a moderately dense jungle or forest foliage shall be demonstrated (see 6.4.2). After the signal has penetrated a simulated jungle, it shall produce the desired display.

4.6.2.4 Jolt test. The signal shall be subjected to 1,750 impacts in three (3) axially perpendicular orientations. These impacts shall consist of half sine wave pulses having 230 ±34.5g peak acceleration for 2.0 ±0.2 milliseconds at a pulse rate of 35 impacts/ minute. After the jolt test, the signal shall then be tested as specified in 4.6.2.1.

4.6.2.5 Jumble test. The signal shall be subjected to random impacts imparted by free-fall inside a rotating, closed, wood (or suitable substitute) material-lined metal box. The lining shall have a minimum thickness of 1/4 inch. The box shall be rotated at a speed of 30 ±2 rpm for a total of 3600 ±10 revolutions. After the jumble test, the signal shall then be tested as specified in 4.6.2.1.

4.6.2.6 Forty foot drop test. The signal shall be subjected to a free fall drop onto a steel plate from a height of 40-feet measured from the lowest point of the test item to the plate. The signal shall then be tested as specified in 4.6.2.1.

4.6.2.7 Altitude test. The signal shall be subjected to an altitude of 50,000 feet for five (5) minutes. The signal shall then be tested as specified in 4.6.2.1.

4.6.2.8 Water immersion test. The signal shall be immersed in water to a minimum depth of 11-inches for a minimum of three (3) hours at a temperature of 70° ±10°F. The signal shall then be tested as specified in 4.6.2.1.

4.6.2.9 Firing pin force (launcher) test. The launcher shall be actuated by hand, with firing pin fully cocked, and using a test plug to simulate the signal but allowing the measurement and recording of the firing pin force (see 6.4.3).

4.6.3 Interchangeability test. The interchangeability of parts having the same part number within the same lots shall be demonstrated.

4.6.4 Environmental conditions tests.

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4.6.4.1 Temperature and humidity tests. The signals and launchers (quantities of each as determined in accordance with 4.3.1) shall be subjected to one 14-day series of temperature and humidity cycles in which the signals are cycled back and forth between -65° and +160°F with a relative humidity of 95% (see 6.4.4). At the conclusion of the 14-day temperature and humidity series, 50% of the signals and launchers shall be conditioned at +160° ± 5°F for a minimum of five (5) hours; the other 50% of the signals and launchers shall be conditioned at -65° ±5°F for a minimum of five (5) hours.

4.6.4.1.1 Low temperature test. The signals and launchers that were conditioned as specified in 4.6.4.1 at -65° ±5°F shall be fired as specified in 4.6.2.1 within 30-seconds after removal from conditioning (signals only). An individual temperature-soaked launcher can be used to fire signals in close sequence for a maximum of 10-minutes without being returned to the temperature conditioning.

4.6.4.1.2 High temperature test. The signals and launchers that were conditioned as specified in 4.6.4.1 at 160° ±5° shall be fired as specified in 4.6.2.1 within 30-seconds after removal from conditioning (signals only). An individual temperature-soaked launcher can be used to fire signals in close sequence for a maximum of 10-minutes without being returned to the temperature conditioning.

4.6.4.2 Salt spray (launcher) test. The launcher shall be subjected to salt spray for 50-hours with a minimum salt concentration of five (5)%. Immediately following the 50-hour salt spray bath, the launcher shall be rinsed in clean water and shall then fire a minimum of seven (7) consecutive signals successfully as specified in 4.6.2.1 without failure and requiring any maintenance. Wiping off visible salt residue with a cloth in the collet area is allowable.

4.6.4.3 Storage tests.

4.6.4.3.1 Extreme high temperature storage test. The signal shall be subjected to an extreme high temperature storage test in an enclosed chamber that shall be maintained at 200° ±5°F for 14-days. At the conclusion of this storage period, the signal shall be tested as specified in 4.6.2.1 after allowing signal to return to ambient temperature.

4.6.4.3.2 High temperature storage test. The signal shall be subjected to a high temperature storage test at 160° ±5°F for 28-days. At the conclusion of this storage period, the signal shall then be tested as specified in 4.6.2.1 within 30-seconds of removal from the conditioning chamber.

4.6.4.3.3 Low temperature storage test. The signal shall be subjected to a low temperature storage test at -65° ±5°F for 28-days. At the conclusion of this storage period, the signal shall then be tested as specified in 4.6.2.1 within 30-seconds of removal from the conditioning chamber.

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4.6.4.4 Vibration tests. The signal kit shall be vibrated at 10 to 60 Hz with an amplitude maintained at 0.10 ±0.01 in. double amplitude up to and including 20 Hz, and at 2 ±0.2g peak for the remaining frequency coverage to simulate transportation (see 6.4.5). After the completion of the vibration test, the signal shall be immersed in 11 to 13 inches of water for one (1) hour. The signal shall then be tested as specified in 4.6.2.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel will be performed by DoD personnel, these personnel will contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements will be maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The signals covered by this specification are intended to be used by aircrew personnel to signal their location as an aid to search and rescue. The signals are intended to penetrate moderately dense jungle foliage and are intended for worldwide application under all anticipated temperature extremes.

6.2 Acquisition requirements. The signals covered by this specification are classified as Common Fireworks, Class C Explosives in accordance with the Code of Federal Regulations 49 CFR 171-178. The contract must invoke Special Packaging Instruction (SPI) F 00-490-7362. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced. (see 2.3)

c. If first article is required. (see 3.1)

d. Item identification instructions. (see 3.6)

e. Packaging requirements. (see 5.1)

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6.3 Subject term (key word) listing.

Burn time
Candle
Candlepower
Display
Penetration test
Rocket motor
Shelf/service life

6.4 Suggested test procedures.

6.4.1 Candlepower and color test. A sample of signals is disassembled. The candle of each sample is mounted on a test fixture, positioned vertically with the ignition end up, and the starter composition ignited. The dominant wavelength, purity, and peak and average luminous intensity of the display are determined.

6.4.2 Penetration test. One successful method of simulating a moderately dense jungle or forest foliage is to set up a fixture consisting of three exterior, commercial grade plywood panels spaced 3.5 to 3.75 inches apart, place this fixture a minimum of 25-feet from the launch site, and fire the signal into the fixture. Each of the three (3) plywood panels are four (4) feet long by four (4) feet wide by 0.25 inch thick. The signal is oriented for firing toward the approximate center of the panels in the fixture.

6.4.3 Firing pin force (launcher). This procedure has been successfully accomplished with a suitable test plug that will simulate the forces required to initiate the primer of the actual signal. With the test plug installed in the launcher and the top of the test plug 0.125 inch from a self-locking force gage (catch gage), the fully cocked firing pin is released, and the reading is recorded. The launcher is actuated by hand.

6.4.4 Temperature and humidity test. This procedure has been successfully accomplished by alternately exposing the signals to temperatures of -65° and +160ºF. MIL-STD-331 offers two (2) methods for performing this cycling. In the two (2) chamber method, items are moved from a cold to a hot chamber and back. The schedule for the two (2)-chamber method is shown in TABLE V. In the single-chamber method, the temperature within the chamber is changed. The schedule for the single-chamber method is shown in TABLE VI. In both cases, note that the schedules call for extended periods at -65° and +160ºF. Both test methods have been modified for a minimum temperature of -65°F.

6.4.5 Vibration test. This procedure has been successfully accomplished using the sweep method and discrete step method. Both methods are discussed in detail in MIL-STD-331 appendix B, paragraph B1.6. *Alternate and Optional Tests*.

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6.5 Definition of hazardous manner. A signal and/or launcher is considered as functioning in a hazardous manner if the signal is held captive in the launcher while the rocket motor burns, the signal explodes, deflagrates, or components separate or come apart within 15-feet of the launcher, or any other abnormality that potentially could cause the operator injury.

6.6 Changes from previous issue. Marginal notations have been used in this revision to identify content changes with respect to the previous issue. Notations were not used for minor editorial changes (format, punctuation, spelling, grammatical, etc.)

TABLE I Burn time and apogee requirements.

Condition Title	Condition Requirement	Display Apogee Minimum & Maximum (feet)	Minimum Burn Time
Ambient	None	600 to 1,600	7.0
Low Temperature	-65° ±5°F	400 to 1,600	4.0
High Temperature	160° ±5°F	600 to 1,600	4.0
High Temp Storage	After storage at 160° ±5°F for a minimum of 28-days	600 to 1,600	4.0
Low Temp Storage	After storage at -65° ±5°F for a minimum of 28-days	400 to 1,600	4.0

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TABLE II First article tests.

TEST PARAGRAPH	SAMPLES TO BE TESTED	REQUIREMENT	REQUIREMENT PARAGRAPH	ACCEPTANCE CRITERIA
4.6.1	271 Signals 1/ 30 Launchers	Workmanship	3.3	Ac 0 Re 1
4.6.1	271 Signals 30 Launchers	Item Identification	3.6	Ac 0 Re 1
4.6.2.2	15 Candles 2/	Burn Time Candlepower Color	3.5.2.1 3.5.2.2 3.5.2.3	Ac 0 Re 1 Ac 0 Re 1 Ac 0 Re 1
4.6.2.9	30 Launchers	Firing Pin Force	3.5.10	Ac 0 Re 1
4.6.4.2	10 Launchers 3/	Salt Spray	3.8.2 3.5.3 3.5.1 a-d	Ac 0 Re 1 Ac 0 Re 1 Ac 0 Re 1
4.6.2.3	10 Signals	Penetration	3.5.4 3.5.3 3.5.1	Ac 4 Re 5 Ac 0 Re 1 4/
4.6.2.4	13 Signals	Jolt	3.5.5 3.5.3	Ac 0 Re 1 Ac 0 Re 1
4.6.2.5	13 Signals	Jumble	3.5.6 3.5.3	Ac 0 Re 1 Ac 0 Re 1
4.6.2.6	5 Signals	40-foot Drop	3.5.7 3.5.3 3.5.1	Ac 0 Re 1 Ac 0 Re 1 Ac 1 Re 2
4.6.2.7	40 Signals	Altitude	3.5.8 & 3.5.3 3.5.1	Ac 0 Re 1 Ac 1 Re 2
4.6.2.8	40 Signals	Water Immersion	3.5.9 & 3.5.3 3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 0 Re 1 Ac 2 Re 3
4.6.4.1 4.6.4.1.1 4.6.4.1.2 5/	40 Signals 5 Launchers	Temperature & Humidity	3.8.1 3.5.3 3.5.1	Ac 0 Re 1 Ac 0 Re 1 Ac 3 Re 4
4.6.4.3.1	10 Signals	Extreme High Temperature Storage	3.8.3.1 & 3.5.3	Ac 0 Re 1
4.6.4.3.2	30 Signals	High Temperature Storage	3.8.3.2 & 3.5.3 3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 0 Re 1 Ac 2 Re 3
4.6.4.3.3	30 Signals	Low Temperature Storage	3.8.3.3 & 3.5.3 3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 0 Re 1 Ac 2 Re 3
4.6.4.4	40 Signals	Vibration	3.8.4 & 3.5.3 3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 0 Re 1 Ac 3 Re 4

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NOTES:

1/ These same signals and launchers shall be used for all following inspections and tests.

2/ Only the candle portion of the signal is required for this test. These samples are in addition to the 271 full up signals. Batch testing acceptable with minimum of two (2) samples per batch and 15-samples minimum per FAT sample.

3/ Each launcher shall fire a minimum of seven (7) signals from subsequent tests (additional conditioning may be required).

4/ Signals shall be required to only produce a display as part of the penetration test.

5/ Twenty signals and two (2) launchers shall be tested in accordance with 4.6.4.1.1 and 20-signals and three (3) launchers shall be tested in accordance with 4.6.4.1.2. Ac/Re criteria are for the entire 40-signal sample.

TABLE III Conformance sample tests.

REQUIREMENT	TEST PARAGRAPH	SAMPLE SIZE	REQUIREMENT PARAGRAPH	ACCEPTANCE CRITERIA	CLASS OF DEFECT
Workmanship	4.6.1	10 Kits 1/	3.3	Ac 0 Re 1	Minor
Burn Time Candlepower	4.6.2.2	5 Candles 2/	3.5.2.1 3.5.2.2	Ac 0 Re 1 Ac 0 Re 1	Major Major
Temperature & Humidity	4.6.4.1 4.6.4.1.1 4.6.4.1.2	20 Signals 3/	3.8.1 3.5.3 3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 0 Re1 Ac 0 Re 1 Ac 2 Re3	Major Major Critical Minor
Water Immersion	4.6.2.8	10 Signals	3.5.9 & 3.5.3 3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 0 Re 1 Ac 2 Re 3	Major Critical Minor
Function	4.6.2.1	40 Signals	3.5.1 a,b,d-f 3.5.1 c,g-i	Ac 0 Re 1 Ac 2 Re 3	Major Minor

NOTES:

1/ Launchers shall each be used to function test seven (7) signals each.

2/ Batch testing is acceptable with minimum of two (2) samples per batch and five (5) samples minimum per lot.

3/ Ten signals shall be tested in accordance with 4.6.4.1.1 and 10-signals shall be tested in accordance with 4.6.4.1.2. Ac/Re criteria are for entire 20-signal sample.

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TABLE IV Requirements cross-reference matrix.

REQUIREMENTS	VERIFICATION
3.1	4.2
3.2	4.6.1
3.3	4.6.1
3.4	4.6.1 & 4.6.4.2
3.5.1	4.6.2.1
3.5.2.1	4.6.2.2.1
3.5.2.2	4.6.2.2.2
3.5.2.3	4.6.2.2.3
3.5.3	4.6.2.1
3.5.4	4.6.2.3
3.5.5	4.6.2.4
3.5.6	4.6.2.5
3.5.7	4.6.2.6
3.5.8	4.6.2.7
3.5.9	4.6.2.8
3.5.10	4.6.2.9
3.7	4.6.3
3.8	4.6.4
3.8.1	4.6.4.1
3.8.1.1	4.6.4.1.1
3.8.1.2	4.6.4.1.2
3.8.2	4.6.4.2
3.8.3.1	4.6.4.3.1
3.8.3.2	4.6.4.3.2
3.8.3.3	4.6.4.3.3
3.8.4	4.6.4.4

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TABLE V Two-chamber method test schedule.

Day	Time Start	Stop	Temperature (degrees ° Fahrenheit)	Relative Humidity
1	0800	0900	Room	
1	0900	1500	-65°	
1	1500	1600	Room	
1-2	1600	0800	160°	95%
2	0800	0900	Room	
2	0900	1500	-65°	
2	1500	1600	Room	
2-3	1600	0800	160°	95%
3	0800	0900	Room	
3	0900	1500	-65°	
3	1500	1600	Room	
3-4	1600	0800	160°	95%
4	0800	0900	Room	
4	0900	1500	-65°	
4	1500	1600	Room	
4-5	1600	0800	160°	95%
5	0800	0900	Room	
5-8	0900	0900	-65°	
8	0900	1500	-65°	
8	1500	1600	Room	
8-9	1600	0800	160°	95%
9	0800	0900	Room	
9	0900	1500	-65°	
9	1500	1600	Room	
9-10	1600	0800	160°	95%
10	0800	0900	Room	
10	0900	1500	-65°	
10	1500	1600	Room	
10-11	1600	0800	160°	95%
11	0800	0900	Room	
11	0900	1500	-65°	
11	1500	1600	Room	
11-12	1600	0800	160°	95%
12	0800	0900	Room	
12	0900	1500	-65°	
12	1500	1600	Room	
12-15	1600	0800	160°	95%

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TABLE VI Single-chamber method test schedule.

Day	Time Start	Stop	Temperature (degrees° Fahrenheit)	Relative Humidity
1	0800	0900	Room	
1	0900	1100	Transition	
1	1100	1600	-65°	
1	1600	1900	Transition	
1-2	1900	0800	160°	95%
2	0800	1100	Transition	
2	1100	1600	-65°	
2	1600	1900	Transition	
2 -3	1900	0800	160°	95%
3	0800	1100	Transition	
3	1100	1600	-65°	
3	1600	1900	Transition	
3-4	1900	0800	160°	95%
4	0800	1100	Transition	
4	1100	1600	-65°	
4	1600	1900	Transition	
4-5	1900	0800	160°	95%
5	0800	1100	Transition	
5-8	1100	0800	-65°	
8	0800	1600	-65°	
8	1600	1900	Transition	
8-9	1900	0800	160°	95%
9	0800	1100	Transition	
9	1100	1600	-65°	
9	1600	1900	Transition	
9-10	1900	0800	160°	95%
10	0800	1100	Transition	
10	1100	1600	-65°	
10	1600	1900	Transition	
10-11	1900	0800	160°	95%
11	0800	1100	Transition	
11	1100	1600	-65°	
11	1600	1900	Transition	
11-12	1900	0900	160°	95%
12	0800	1100	Transition	
12	1100	1600	-65°	
12	1600	1900	Transition	
12-15	1900	0900	160°	95%

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TABLE VII Examination of product.

No.	Defects	Critical	Major	Minor
1	Canted nozzle at rear of signal omitted 1/	X		
2	Nozzle of signal incorrectly installed 1/	X		
3	Nozzle clogged with foreign matter 2/	X		
4	Launcher missing	X		
101	Signal dented 1/		X	
102	Loose actuation knob, collet, or handle of launcher 1/		X	
103	Firing pin interfering with collet 1/		X	
104	Color of signal assembly not as specified 3/		X	
105	Diameter of unpainted surface of signal not as specified 1/		X	
106	Primer of signal omitted or incorrectly located 1/		X	
107	Signal falls from collet when launcher is held with distal end pointed down		X	
108	Bandoleer containing less than 7 signals		X	
201	Instruction sheet missing			X
202	Knurling omitted from launcher or lanyard of launcher missing 1/			X
203	Burrs on launcher 1/			X
204	Plastic bag torn			X
205	Launcher not tied to bandoleer			X

NOTES:
1/ Inspection for this defect shall be performed prior to kit assembly.

2/ Nozzle holes shall be visually inspected by using a minimum three (3)-power magnifying glass.

3/ This inspection shall pertain to the color of the signal only, for example, to determine that the signal is painted red when a red (display) signal is specified and not to determine the color shade of the signal.

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TABLE VIII Classification of defects.

No.	Defects	Critical	Major	Minor
5	Launcher holds signal captive while rocket motor burns; signal explodes, deflagrates, or burns within 15 feet of launcher; or rocket motor vents and snuffs out propellant system	X		
6	Components of signal separate or come apart within 15 feet of launcher	X		
7	Color of candle display not in accordance with color identification of signal	X		
8	Apogee of signal deviates more than 45 degrees form vertical	X		
109	Maximum height of display less than 400 feet for signal that has been subjected to low temperature conditioning		X	
110	Maximum height of display less than 600 feet or more than 1,600 feet		X	
111	Maximum height of display exceeds 1,600 feet after signal has been subjected to high temperature conditioning.		X	
112	Launcher requires maintenance before launcher fires minimum of 7 signals		X	
113	Display fails to meet burn time and candlepower requirements of 3.5.2.1 and 3.5.2.2.		X	
114	Apogee of display deviates more than 30 degrees but less than 45 degrees from vertical		X	
115	Burn time of display less than seven (7) seconds for red signal (not conditioned) 1/		X	
116	Burn time of display less than four (4) seconds for red signal that has been subjected to humidity or high and low temperature conditioning		X	
206	Display hitting ground while candle is still burning			X
207	Candle ignites below 400 feet			X

NOTES:

1/ Non-ignition of the display shall be considered zero burn time.

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CONCLUDING MATERIAL

Custodians:
Air Force – 170
Army – GL

Preparing activity:
Air Force – 170

Project 1370-2018-002

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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